

IsoConserve: Statement History of the Conservation Laws

Companion note to “The Isomorphic Scheduler”

Fabian Franz
Isomorphic AI

Claude (Anthropic)
Research Collaborator

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This note conserves the evolution of the law statements across the series and their Lean 4 formalization. Nothing here is required to use the theory; it is the ledger of how the statements reached their final form.

1 L2: from prose to proof

Original statement (series papers 04 and 08). *“In any interval with no progress, convertible stock is monotone non-increasing.”* The prose carried a parenthetical—“in the no-injection regime that holds once a set is blocked”—doing load-bearing work without being part of the claim.

The counterexample the machine forced. A no-progress interval may contain a *flow* step: reservoir shares injected into a runnable process that then fails to convert. Convertible stock rises; no progress is made. The unrestricted statement is false.

Corrected statement (proven).

Law 1 (L2, final form). *Along any reachable sequence of non-injecting transitions (drain and yield steps), convertible stock is monotone non-increasing. Moreover, each blocked process’s stock is individually non-increasing under arbitrary mixed sequences, and flow steps leave blocked stocks unchanged.*

Why nothing downstream weakens. Detection reads blocked processes; their stocks are monotone unconditionally (`blocked_stock_monotone` in `Reachable.lean`), so detection soundness (`detection_sound`) is untouched.

The lesson, recorded. The parenthetical was the hypothesis. Formalization did not merely check the theory; it falsified one clause and returned the repaired clause with a proof—the epistemic invariant applied to the theory itself: let the ledger overrule the narrative.

2 Provenance

Original statements: series papers 04 (§4.2) and 08-expanded (§4.4), repository `isomorphic-ai/scheduler`. Corrected forms: `IsoConserve/L2Monotone.lean` and `IsoConserve/Reachable.lean`, branch `iso-conserve-lean`. The prose papers retain their original wording pending revision; this note is the authoritative reconciliation until then.